

# Christopher Farber

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## EDUCATION

- Purdue University** Fall 2026 (Expected)  
*Master of Science in Computer Science* West Lafayette, IN  
• **Selected Coursework:** Statistical Machine Learning, Natural Language Processing, Computer Vision with Deep Learning
- Purdue University** Fall 2025  
*Bachelor of Science in Computer Science (3.81 GPA)* West Lafayette, IN  
• **Selected Coursework:** Systems Programming, Analysis of Algorithms, Data Mining & ML, Intro to AI

## PROFESSIONAL EXPERIENCE

- Purdue University** September 2025 - Present  
*Graduate Research Assistant* West Lafayette, IN  
• Applying ML methods to predict mortality outcomes in DMD patients using CMR data, in collaboration with the NIH.  
• Developing and evaluating UQ methods for multi-fidelity neural operator models, collaborating with Sandia National Labs.  
• Responsible for model development and experimental analysis under supervision of Dr. Guang Lin, Purdue University.
- Kohl's** June 2025 - August 2025  
*Software Engineering Intern* Menomonee Falls, WI  
• Engineered a full-stack React/TypeScript + Kotlin/Spring Boot application to streamline the creation, sharing, and execution of test suites, significantly reducing test setup time for Kohl's Technology teams.  
• Upgraded 4 production-level repositories to use the new version of Kohl's' GitLab DevOps pipeline.  
• Delivered a workshop on Framer Motion to an audience of 40+ people, showing how it can enhance the Kohl's landing page.
- iRODS** June 2024 - August 2024  
*Software Engineering Intern* Chapel Hill, NC  
• Refactored React app for the Zone Management Tool to communicate with newly-developed HTTP API.  
• Developed a TypeScript wrapper library for the HTTP API, consisting of 120+ types and 60+ unit tests written with Jest.
- National Transportation Consultants** June 2023 - August 2023  
*Software Engineering Intern* Fishers, IN  
• Built an Azure AI model to detect and extract fields from 10+ document types, achieving 2x processing speed.  
• Coordinated with CEO and software development team to integrate model into company's operations.
- Primary Record** June 2022 - August 2022  
*Software Engineering Intern* Fishers, IN  
• Collaborated with company founders to develop a high-fidelity proof of concept for an OCR-based prescription storage feature, using React and GCP, establishing the foundation for its eventual integration into the flagship application.

## PROJECTS

- Geospatial ML Group** | *Python, PyTorch, GIS* August 2024 - Present  
• Trained a ConvLSTM model with PyTorch to predict levels of PM2.5 and correct the bias of NASA's GEOS-FP model.  
• Collaborating with 2 NASA scientists to integrate model into the MAIA satellite project.  
• Established a data to training pipeline, enabling group to get a headstart on model analysis.  
• Presented poster at Purdue's 2024 GIS Day conference.
- Project oCEANIC** | *Python, Flask, Redis, AWS S3, Next.js, Leaflet.js* January 2024 - Present  
• Developed agent-based seafaring model using archaeological and paleoclimate data, revealing patterns of historical seafaring.  
• Preprocessed and saved 40 years of hourly NetCDF wind data from NASA's MERRA-2 dataset into an AWS S3 bucket.  
• Created and deployed Next.js website that allows users to visualize, interact with, and export simulations from the model.  
• Deployed Flask server that runs simulations for the website, along with a Redis server to store simulation data.  
• Presented preliminary research results at a workshop at the University of Hawaii Hilo, research talks at Purdue's 2024 & 2025 Fall Undergraduate Research Conferences, and posters at Purdue's 2024 & 2025 GIS Day Conferences.
- Analysis of Online HPV Vaccine Hesitancy** | *Python, Reddit API, pandas, NLTK* September 2023 - May 2024  
• Created a script to automatically collect data via Reddit's API from HPV-related subreddits each day, totaling in 316 posts.  
• Utilized scikit-learn to vectorize the data with TF-IDF, then K-Means and TSNE/SVD to find 3 significant clusters.  
• Applied NLP and statistical methods to analyze differences in sentiment and subjectivity across clusters, via NLTK.  
• Presented poster at Purdue's 2024 Spring Undergraduate Research Conference.

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, JavaScript, TypeScript, Kotlin, R  
**Frameworks:** Node.js, Next.js, Tailwind CSS, Jest, Cypress, Flask, Material-UI, PyTorch, Spring Boot  
**Developer Tools:** Git, Google Cloud Platform, Microsoft Azure, AWS S3, Docker, nginx, CI/CD, Redis, Postgres  
**Libraries:** React.js, Axios, pandas, NumPy, Matplotlib, TensorFlow, Keras, scikit-learn, NLTK, SpaCy